```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

```
df_train = pd.read_csv("train_KNN.csv")
df_test = pd.read_csv("test_KNN.csv")

df_train.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Next steps: Generate code with df_train New interactive sheet

```
df_train["Age"] = df_train["Age"].fillna(df_train["Age"].median())
df_test["Age"] = df_test["Age"].fillna(df_test["Age"].median())

df_train = df_train.dropna(subset=["Embarked"])
df_test = df_test.dropna(subset=["Fare"])

df_train = df_train.drop(["Cabin", "Name", "Ticket", "PassengerId"], axis=1)
df_test = df_test.drop(["Cabin", "Name", "Ticket", "PassengerId"], axis=1)
```

```
df_train = pd.get_dummies(df_train, drop_first=True)
df_test = pd.get_dummies(df_test, drop_first=True)
```

```
X_train = df_train.drop("Survived", axis=1)
y_train = df_train["Survived"]

X_test = df_test.drop("Survived", axis=1)
y_test = df_test["Survived"]
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

knn_accuracy = accuracy_score(y_test, y_pred)
print("KNN Accuracy:", knn_accuracy)
```

```
KNN Accuracy: 0.8129496402877698
```

```
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

```
lr_accuracy = accuracy_score(y_test, y_pred_lr)
print("Logistic Regression Accuracy:", lr_accuracy)
```

```
Logistic Regression Accuracy: 0.9424460431654677
```

```
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)

dt_accuracy = accuracy_score(y_test, y_pred_dt)
print("Decision Tree Accuracy:", dt_accuracy)
```

```
Decision Tree Accuracy: 0.7961630695443646
```

```
print("KNN Accuracy:", knn_accuracy)
print("Logistic Regression Accuracy:", lr_accuracy)
print("Decision Tree Accuracy:", dt_accuracy)
```

```
KNN Accuracy: 0.8129496402877698
Logistic Regression Accuracy: 0.9424460431654677
Decision Tree Accuracy: 0.7961630695443646
```

Close

```
## Findings and Interpretation

In this experiment, three machine learning algorithms were
applied to the Titanic dataset to predict passenger survival.

The K-Nearest Neighbors (KNN) model achieved an accuracy of
approximately 81%, showing moderate performance. Logistic
Regression performed the best with an accuracy of around 94%,
indicating that it is the most suitable model for this dataset.
The Decision Tree model achieved around 79% accuracy, which is
slightly lower and may be affected by overfitting.

From the comparison, it is clear that Logistic Regression
provides the most reliable results, while KNN and Decision Tree
show comparatively lower performance.

This experiment demonstrates that different machine learning
models produce different results, and choosing the right model is
important for achieving high accuracy.
```

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